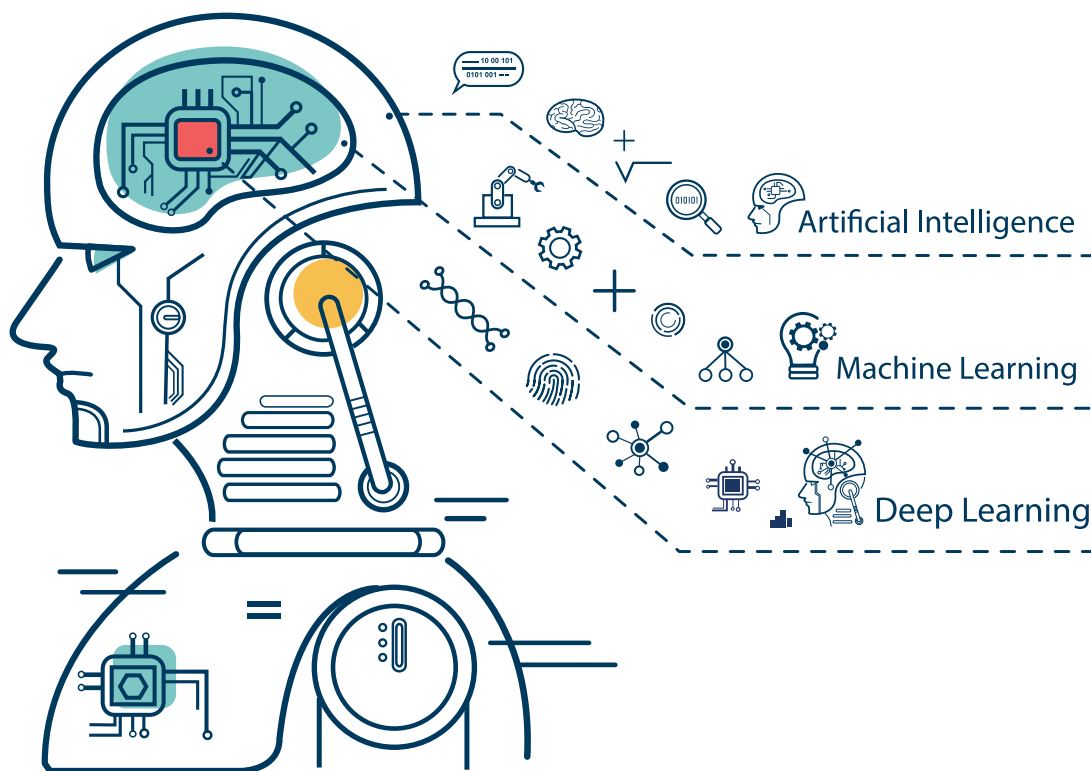


AI, ML and DL: What's the Difference?

We often use the terms artificial intelligence, machine learning and deep learning interchangeably, even though we read or hear about them almost each day. This article explains how these technologies evolved and in what ways they differ.



Artificial intelligence (AI), machine learning (ML), and deep learning (DL) are often used interchangeably; however, they are not quite the same things. AI is the broadest concept of all, and gives a machine the ability to imitate human behaviour. ML is the application of AI into a system or machine, which helps it to self-learn and improve continually. Lastly, DL uses complex algorithms and deep neural networks to repetitively train a specific model or pattern.

Let's look at the evolution and journey of each term to get a better

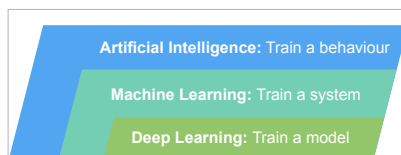


Figure 1: Overview of AI, ML and DL

understanding of what AI, ML and DL actually refer to.

Artificial intelligence

AI has come a long way since the last 70-odd years, infiltrating into every aspect of our life, whether we know it, and like it or not. Advancements

in machine learning and deep learning over the last decade have created an AI boom across industries and organisations of all sizes. Cloud service providers have added to the momentum by developing open source services that are available for free and by offering new use cases.

AI is perhaps the most worked upon concept since 1956. By 2015, the wide availability of GPUs made parallel processing faster, powerful and cheaper. Cheaper options led to humongous storage of Big Data (plain text to images, to mapping, etc). This